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The Hidden Journey Of Pearls

Pearls are unique gemstones that get created by living organisms called oysters. “They don't start out perfect, they begin with an injury, a grain of sand or a parasite slips into the soft body of an oyster, the oyster doesn't like it because it feels irritating. It's a wound but instead of rejecting it the oyster slowly secretes layers of aragonite and conchiolin.” This creates the material called nacre and it's also known as mother-of-pearl, it encases the irritant and protects the mollusc from it. Over time the oyster keeps adding more and more layers of nacre. After a while, all these layers build up and form a pearl. Pearls are harvested by people who raise or collect oysters from the ocean. Farmers take careful care of the oysters for several years or months while the pearls slowly form inside. During this time, they keep it in clean water and protected from harm so they can grow properly. Once the oysters are ready they are brought up from the water and carefully opened, the pearl is removed by hand. According to the National Oceanic and Atmospheric Administration pearl farming includes growing the oysters and collecting them when the pearls are fully formed. “Many farm workers try to open the oyster carefully so they can stay alive and possibly produce more pearls.” After harvesting the pearls are cleaned, separated by size and quality then prepared to be sold or made into jewelry. A cultured pearl is formed in the same way as a natural pearl. Cultured pearls are pearls that are made with the help of people. “It involves an expert pearl technician putting a small object like a bead called a nucleus which acts as the irritant to the pearl.” The oyster then is when it covers it with layers of the shiny material nacre

which the pearl forms over time. Even though people start the process, the oyster still makes the pearl naturally. Only about 1 in 10,000 oysters produces one naturally. Because of this, most pearls sold today are cultured pearls. Natural pearls are pearls that form without any help from people. “They are the ones that happen when something small, like a piece of sand or a tiny parasite gets inside an oyster by accident.” Then it's the same process with the nacre. Natural pearls are very rare because they depend on chance, so they are usually harder to find than cultured pearls. Pearls come in many different colors like white, cream, pink, silver, gold, black, blue, and green depending on the type of oyster and water conditions. They are the only gem created by a living animal. Ama are traditional women divers from Japan who have been diving for over 2,000 years and training often began around the age of 12. The word “ama” means “sea women” These women are known for free diving in which they go underwater without tanks or heavy equipment. They go underwater “dive to depths of around 30 feet and can hold their breath for up to two minutes.” and usually wear wetsuits, goggles, sometimes weights to help them go deeper. They usually collect seafood like abalone, seaweed, and shellfish but also played a big role in Japan’s pearl industry. “While ama did not originally dive for pearls, their role changed dramatically in early 1893 when Kōkichi Mikimoto developed the world’s first successful method for culturing pearls” ama helped gather pearl oysters from the ocean and made it possible for pearls to be produced and sold around the world. “Their higher body fat provided insulation against cold seawater, and their calm, steady movements underwater made them efficient free divers” and had to train for years to build strength and control. “In the 1950s, there were more than 17,000 ama divers in Japan. Today, fewer than 2,000 remain.” They would mostly be found in Toba City. However, they are still an important part of Japan’s culture and history especially in coastal areas like Mie Prefecture.



Fig. 1. Ama divers in the water.

The image shows two ama divers in the water near a pearl farm in Japan. They are wearing traditional white diving clothes and are floating next to large wooden tubs that help them rest and carry oysters. In the background rafts can be seen floating on the water. This picture shows how ama divers worked closely with Japan's pearl industry and spent long hours diving to collect oysters from the sea.



Fig. 2. Pearls inside an oyster.

The image shows pearls inside the shell of an oyster. The inside of the shell has a shiny colorful surface called nacre, which is the same material used to make pearls. The pearls seem smooth and round because the oyster slowly covered them with many layers of nacre over time. This picture helps show how pearls grow naturally inside oysters.

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